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PALAEOMAGNETISM OF PERMO-TRIASSIC ROCKS FROM NORTHERN SLOVENIA, YUGOSLAVIA, AND THE EASTERN MARGIN OF THE ADRIATIC PLATE

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ABSTRACT

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In northern Slovenia (Yugoslavia, $\lambda = 15°E$, $\varphi = 46°N$), 41 sites of Permo-Triassic red sandstones and volcanic rocks have been sampled for a palaeomagnetic study. The rock units are situated either in the Southern Alps or in the transition zone between the Southern Alps and the Dinarides. More than half of the sampled sites revealed that the primary remanence was seriously affected by partial or entire magnetic overprint in connection with the Alpine orogeny. This was manifested by either too steep inclinations, or too weak or inconsistent and unstable remanence. Only fourteen sites revealed the typical flat inclinations of Permo-Triassic rocks. A group of six sites in northwestern Slovenia (west of Ljubljana) gave a remanence direction ($N = 6$, $R = 5.431$, $D = 151.4°$, $I = -11.4°$, $k = 8.8$, $\alpha_{95} = 19.3°$) which is in agreement with the mean Permian remanence direction of the Bolzano Rhyolites in the Italian part of the Southern Alps ($D = 148°$, $I = -20°$, after Zijderveld and Van der Voo, 1973), indicating that this zone is still part of the Adriatic plate. However, a group of eight sites in northeastern Slovenia (east of Ljubljana) has a remanence direction ($N = 8$, $R = 6.991$, $D = 225.9°$, $I = -15.0°$, $k = 6.9$, $\alpha_{95} = 18.8°$) which is more in agreement with the central European Permo-Triassic direction of remanence ($D = 204°$, $I = -14°$).

It is therefore concluded that the eastern margin of the Adriatic Plate crosses the region of Ljubljana in an approximately north–south direction.

INTRODUCTION

Various reconstructions of the Western Mediterranean have been made in the recent years by combining geological and palaeomagnetic evidence (Dewey et al., 1973; Hsü, 1977; Channell et al., 1979; Vandenberg, 1979). A common feature was in all cases the assumption of an Adriatic microplate. The extensions of this microplate are roughly given in the reconstruction by Hsü (1977).

For the determination of the eastern margin of the plate only few palaeomagnetic data are available sofar from the Balcan region (Krs and Roth, 1979; Márton and

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Márton, 1978, 1981) which could contribute to this problem.

In the Italian part of the Southern Alps, the Permian and Triassic rocks south of the Periadriatic line have northwestern–southeastern declinations of the characteristic remanent magnetization (Zijderveld and Van der Voo, 1973). These directions are regarded as typical for the Adriatic plate for this time interval, regardless of whether the Adriatic plate is considered as a separate unit or as a promontory of the African plate (Channell et al., 1979).

We expected to encounter similar northwestern–southeastern declinations also in the Yugoslavian part of the Southern Alps in the region south of the Periadriatic Line. The aim of this investigation was to examine how far to the east these typical "Adriatic" Permo-Triassic directions of remanence can still be observed, thus yielding palaeomagnetic constraints for the northeastern margin of the Adriatic plate.

GEOLOGY AND SAMPLING

The area where the samples have been taken belongs to the Slovenian part of the Southern Alps, as well as to the transition zone between the Southern Alps and the Dinarides. In contrast to the Eastern Alps north of the Periadriatic line with extensive folding and nappe-tectonics as well as young metamorphism, the Southern

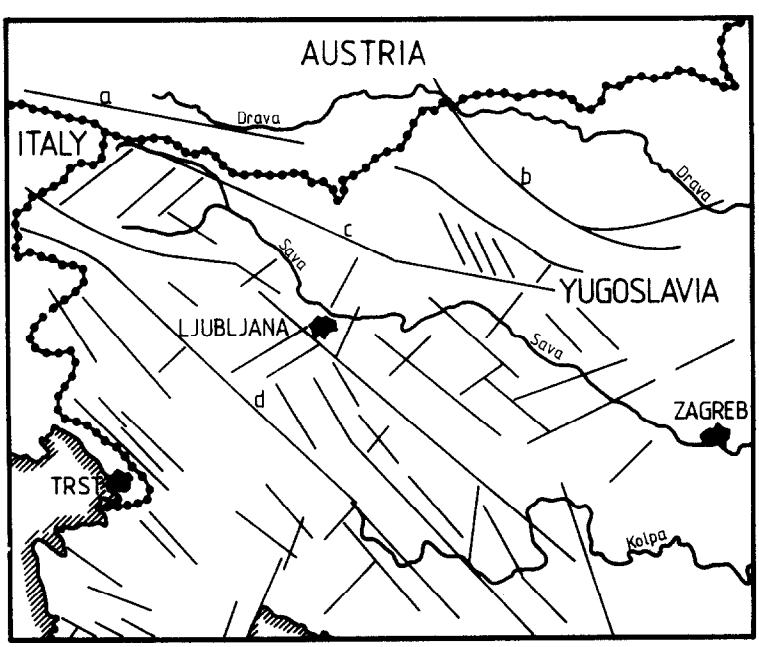


Fig. 1. Main fracture zones in northern Slovenia. Redrawn from Gospodarić (1970), yet simplified. *a* = Periadriatic line; *b* = Donati fracture zone; *c* = Sava fracture zone; *d* = Idrija fracture zone.

TABLE I

Schematic stratigraphy and age relationships of the investigated rocks

Formation	Stage	Lithology	Sampling
Jurassic	Lias	Limestones	
	Rhaetian	Dachsteinkalk	
	Norian	and dolomite	
	Carnian	Raibler layers	bauxites, sandstones
Triassic	Ladinian	Platy limestone, marls	volcanic rocks
	Anisian	Dolomite	
	Scythian	Dolomite, limestone, shales	
	Thuringian	Bellerophon-layers	
Permian	Saxonian	Gröden layers Tarvisio breccia Trogkofel limestone	sandstones
	Autunian	Rattendorf layers	
Carboniferous		Sandstones, limestones, shales	

Alps are characterized by (a) absence of nappes, (b) only small amounts of overthrusting, (c) intensive volcanic activities during the Permian, Triassic and Tertiary, (d) no regional metamorphism. However, these conditions are only valid for the more westerly part of the Southern Alps. Towards the east, especially in the transitional area between the Southern Alps and the Dinarides, large horizontal displacements become frequent as well as folding. There are also some indications of a, however, slight regional metamorphism.

Gospodarić (1970) has summarized the knowledge about the fracture zones in northern Yugoslavia in the area between the Southern Alps and the Dinarides. Along some of these fracture zones, which are shown in Fig. 1, considerable horizontal displacements have been observed. The area suffered also folding, especially in the eastern parts, starting in Upper Cretaceous and ending in Upper Miocene.

The rocks, which have been sampled for this investigations at 36 localities (41 sites) are listed in Table I. They consist of red sandstones, bauxites and volcanic rocks. Figure 2 shows the sampling localities (numbers refer to site numbers) together with the simplified geology (taken from the Geological map of Yugoslavia, 1: 500,000).

TABLE II

List of site numbers, locality name, rock type, stratigraphic age, strike and dip of the formation, number of sampled cores, number of sites at each locality

Locality No.	Locality	Rock type	Age	Strike/dip (degr./degr.)	Number N of samples (cores)	Number of sites
1	Stopnik	diabase	Ladinian	295/50 NE	10	1
2	Lajše	pillow lava	Ladinian	undetermined	10	1
3	Kamna Gorica	keratophyr	Ladinian	210/25 NW	8	1
4	Kokra	keratophyr	Ladinian	200/40 NW	12	2
5	Kamniška Bistrica	keratophyr	Ladinian	264/85 N	7	1
6	Kranjska Reber I	keratophyr tuff	Ladinian	300/40 NE	10	1
7	Kranjska Reber II	keratophyr	Ladinian	140/70 SW	6	1
8	Pirešica	keratophyr	Ladinian	260/40 N	18	1
9	Črnolica	keratophyr	Ladinian	360/35 E	14	1
11	Bohor	keratophyr	Ladinian	200/50 NW	16	3
12	Brsnina	porphyrite	Ladinian	206/85 NW	12	1
13	Tržič	ignimbritic tuff	Ladinian	265/60 N	10	1
14	Sopota	sandstone	Saxonian	14/15 E	10	1
15	Svibno	sandstone	Saxonian	222/28 NW	7	1
16	Štore	keratophyr tuff	Ladinian	40/70 SE	11	1
17	Hrastino	sandstone	Saxonian	251/140 NE (overtilted)	21	1

18	Stična	siltstone	Saxonian	314/119 NE (overtilted)	10	1
19	Žirovski I	sandstone	Saxonian	270/25 N	10	1
20	Žiri	sandstone	Saxonian	192/55 W	11	2
21	Savske Jame	sandstone	Saxonian	180/65 W	13	1
22	Košutnik	sandstone	Saxonian	158/92 SW (overtilted)	12	1
23	Dolžanova Soteska	sandstone	Saxonian	220/35 NW	9	1
24	Podljubelj	siltstone	Saxonian	243/51 NW	11	2
25	Jesenice-Mirca	sandstone	Saxonian	176/45 W	11	1
26	Hrastnik	sandstone	Saxonian	220/53 NW	13	1
27	Jezersko Kozji voh	sandstone	Saxonian	210/32 NW	12	1
28	Vransko	sandstone	Saxonian	312/75 NE	16	1
29	Rečica-Laško	sandstone	Saxonian	332/32 NE	14	1
30	Trnovec	sandstone	Saxonian	148/60 SW	14	1
31	Sovodnje I	sandstone	Saxonian	200/32 NW	20	1
32	Sovodnje II	sandstone	Saxonian	18/75 E	18	1
33	Otaležl I	sandstone	Saxonian	42/40 SE	6	1
34	Otaležl II	sandstone	Saxonian	340/60 NE	20	1
37	Borovnica II	sandstone	Carnian	260/32 N	17	1
38 *	Kokarje	bauxite	Oligocene	uncertain	14	1
40	Žirovski II	sandstone	Saxonian	12/35 E	15	1

* Only NRM was measured. Bedding correction uncertain.

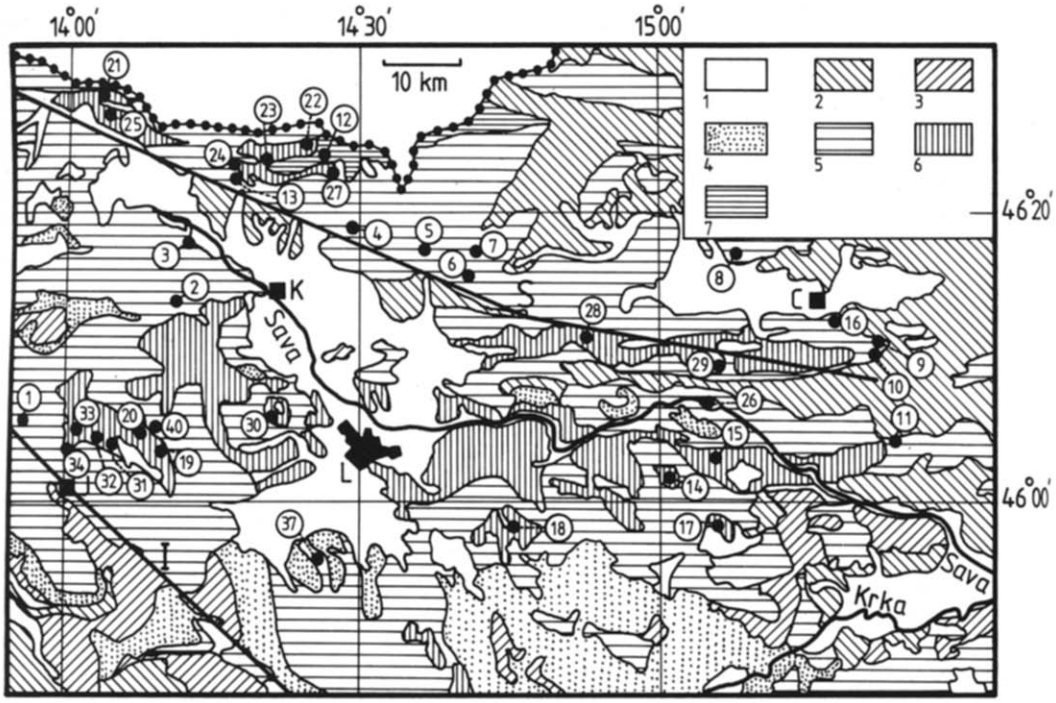


Fig. 2. Simplified geological map of the sampling area, taken from the Geological Map of Yugoslavia (scale 1 : 500,000, issued by the Federal Geological Institute in Belgrade, 1971). The sampling localities are shown as dots, the numbers referring to the site numbers. Bold lines indicate the positions of the Sava (*S*) or Idrija (*I*) fracture zones respectively. *J* = Idrija; *K* = Kranj; *L* = Ljubljana; *C* = Celje. *1* = Quaternary; *2* = Tertiary; *3* = Cretaceous; *4* = Jurassic; *5* = Triassic; *6* = Permian; *7* = Carboniferous.

The position of the main fracture zones are indicated by bold lines (see also Fig. 1).

Table II gives for all sites: site number, locality, rock type, age, strike and dip of the rock unit, number of samples (cores).

In general the rocks had a fresh appearance. However, some indications of an at least low-grade metamorphism could not be overlooked at some sites. The pillow lava and the diabases (sites 1, 2 and 10) were spilitized and showed the formation of secondary chlorite. Some of the keratophyrs were also visibly altered (i.e. sites 5, 6, 9), and similar effects could be observed on many sandstones (formation of secondary mica). The question of a partial or entire remagnetization, which was of minor importance in the palaeomagnetic study of the rocks in the Italian part of the Southern Alps (Zijderveld and Van der Voo, 1973) was serious in our study and became evident already during sampling.

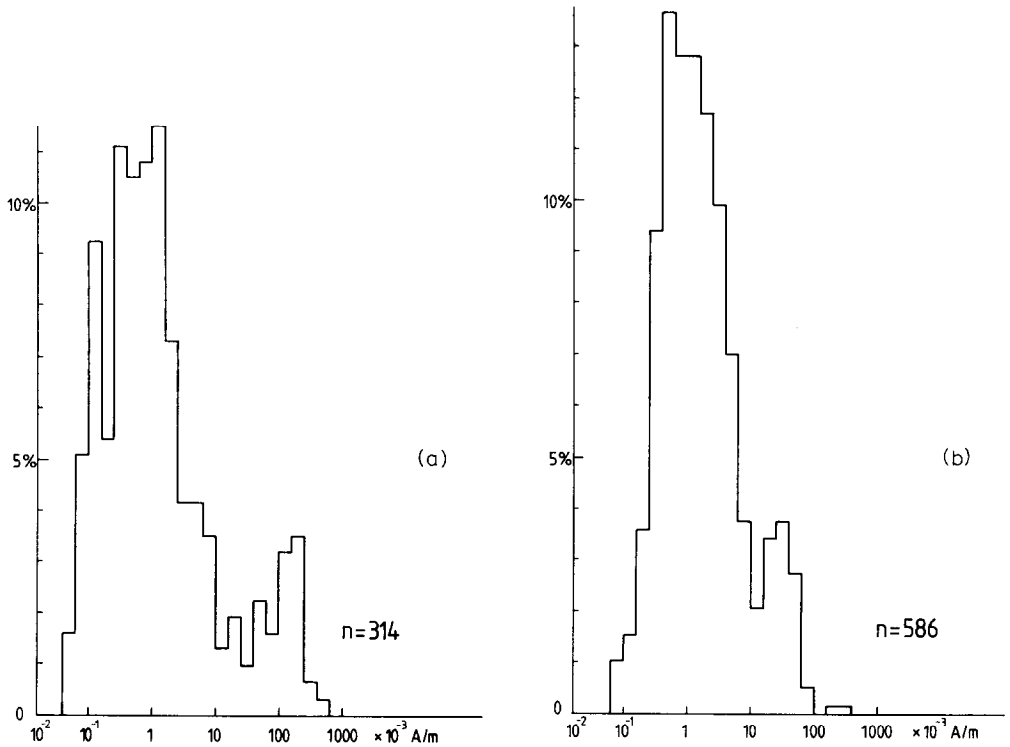


Fig. 3. Histograms of the intensities of the natural remanent magnetization (NRM) in a semi-logarithmic plot. a. Sedimentary rocks (sandstones and bauxite). b. Volcanic rocks (pillow lava, diabase, keratophyrs and their tuffs). n = number of specimens.

NATURAL REMANENT MAGNETIZATION (NRM)

NRM was measured with a shielded Digico spinner magnetometer. Figure 3 shows a histogram of the NRM intensities of the sedimentary (3a) and of the volcanic rocks (3b). The low intensities of most of the volcanic rocks indicate that a secondary chemical remanence (CRM) is more likely the dominant remanence component and not a primary thermoremanence (TRM). The NRM intensities of the sandstones have quite normal values. The maximum sensitivity of our Digico magnetometer (10^{-4} A/m) was just sufficient for the measurement of these rocks. Some of the high NRM intensities of the sedimentary rocks come from the only Oligocene bauxite (site 38), which was not studied in detail and which had only scattered directions of remanence.

Table III is a list of the site mean directions of NRM together with the statistical parameters. The site mean directions (without circles of confidence) are plotted in Fig. 4 in equal area projection. For some sites (4, 11, 20, 24) more than just one

TABLE III

NRM data for each site *

Site number	N	n	R	D (°E)	I (°)	k	α_{95} (°)	Remarks
1	10	8	7.022	280.4	55.5	7.2	18.5	
2	10	9	7.938	23.2	74.0	7.5	17.0	
3	8	8	6.776	37.6	−42.1	5.7	20.7	
4/1	3	3	2.964	118.0	−71.7	56.0	10.8	two groups, 100 m apart
4/2	9	9	8.816	285.4	−19.6	44.1	7.0	
5	7	7	5.806	263.7	−38.9	5.0	23.6	
6	10							scattered
7	6							scattered
8	18	16	14.355	2.2	9.0	9.1	11.6	
9	14	12	10.588	232.6	−70.0	7.8	14.5	
11/1	10	9	7.154	241.7	33.8	4.3	22.4	three groups, 100 m apart
11/2a	3	3	2.957	347.9	48.4	46.2	11.9	
11/2b	3	3	2.619	232.4	47.3	5.3	35.3	
12	12							scattered
13	10	8	7.935	18.2	8.4	107.2	4.8	
14	10							scattered
15	7	7	6.782	283.6	18.2	27.6	10.1	
16	11							scattered
17	21	19	18.260	116.3	14.7	24.3	6.5	
18	10	10	9.675	236.5	12.0	27.7	8.4	
19	10							scattered
20/1	3	3	2.993	88.0	4.8	297.4	4.7	two groups, 50 m apart
20/2	8	8	7.569	130.7	12.3	16.3	12.3	
21	13	13	12.699	83.7	49.7	39.9	6.1	
22	12							scattered
23	9	9	8.570	27.8	64.9	18.6	10.8	
24/1	5	5	4.951	163.7	33.6	82.1	6.9	two groups
24/2	6	6	5.411	264.7	4.5	8.5	19.6	
25	11	11	10.504	284.7	60.6	20.2	9.4	
26	13	11	10.551	239.7	41.7	22.3	8.9	
27	12	11	9.959	283.4	−43.8	9.6	13.6	
28	16	15	14.030	322.9	43.2	14.4	9.5	
29	14	13	9.503	175.2	27.5	3.4	21.0	
30	14	11	8.965	223.1	−41.0	4.9	19.0	
31	20	20	16.482	146.5	64.1	5.4	13.5	
32	18							scattered
33	6	6	5.543	184.3	76.9	10.9	17.3	
34								scattered
37	17	17	16.833	196.5	−25.3	96.1	3.5	
38	14							scattered
40	15	15	13.649	136.1	−3.5	10.4	11.2	

* N = number of cores; n = number of cores used for the computation of the mean direction; R = vector sum; D = declination; I = inclination; k = precision parameter; α_{95} = semiangle of the cone of confidence.

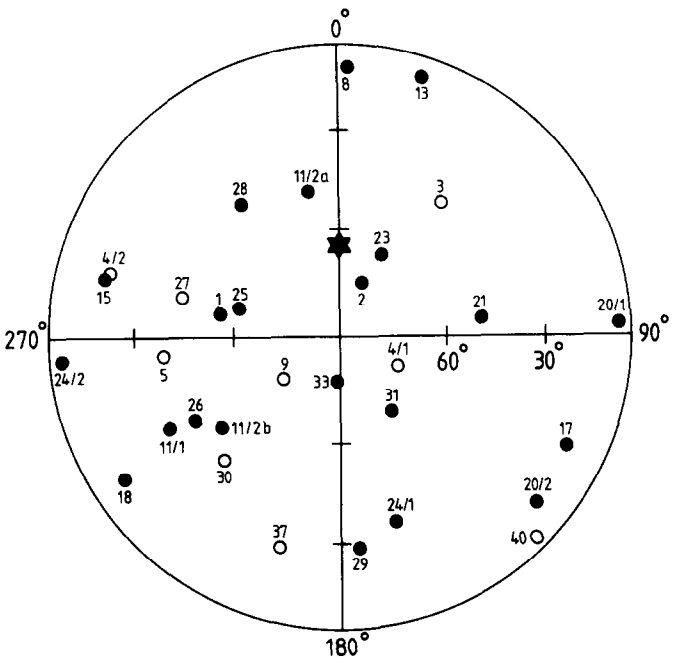


Fig. 4. Site mean directions of the NRM. Due to their large internal scatter the ovals of confidence are not plotted (see Table III). Star: present geomagnetic field in the sampling area. Numbers refer to the site numbers. Open and closed symbols: negative or positive inclination respectively.

group of NRM directions was found. The remanence directions of many sites were extraordinarily scattered (6, 7, 12, 14, 16, 19, 22, 32, 34, 38) and no mean direction has been computed for them.

The mean NRM directions of the sites are also very scattered and show no grouping, neither around the direction of the present geomagnetic field in the sampling area (star in Fig. 4) nor around any other direction or along a great circle. This indicates that viscous remanence components in the direction of the present geomagnetic field seem to be of minor importance in all cases.

DETERMINATION OF THE CHARACTERISTIC REMANENT MAGNETIZATION (ChRM)

This was done in the usual way by entire demagnetization of several pilot specimens from each site. AF demagnetization was applied to the volcanic rocks. The sandstones were thermally demagnetized. The equipment, which was used, consisted of a non-magnetic furnace (for twelve standard specimens) with Helmholtz coils and double μ-metal shielding for the cancellation of the geomagnetic field in the laboratory. The laboratory field could be kept below 30 nT during the demagnetization experiments.

TABLE IV

Characteristic remanence directions before (left) and after (right) bedding corrections. See also the legend of Table III. In the second column information is given about the optimum conditions for AF or thermal demagnetization, respectively. Last column: classification into sites in northwestern (WS) or northeastern (ES) Slovenia, respectively. SI = sites with too steep inclinations. ?: uncertain results.

Site number	mT; °C	Before bedding correction						After bedding correction		
		n	R	D(°E)	I(°)	k	α_{95}(°)	$D_{corr.}$ (°E)	$I_{corr.}$ (°)	Remarks
1	200 mT	7	6.624	16.3	−50.7	15.9	13.3	63.5	−34.8	?
2	50 mT	7	6.694	37.5	70.9	19.6	12.0			see Table II
3	30 mT	7	6.836	133.5	−73.1	36.6	8.8	69.5	−63.6	SI
4/1	30 mT	3	2.969	136.0	−73.8	64.3	10.1	45.6	−54.5	SI
4/2	50 mT	9	8.822	282.9	−15.1	45.0	7.0	294.5	−16.0	?
5	15 mT	7	6.155	183.7	−24.5	7.1	19.9	150.0	−10.9	WS
6	30 mT	8	7.670	151.4	−46.9	21.2	10.8	141.2	−10.6	WS
7	40 mT	6	5.527	228.5	11.9	10.6	17.6	218.3	2.7	ES
8	50 mT	15	14.354	352.0	−25.0	21.7	7.8	8.1	−17.7	ES
9	75 mT	12	11.682	254.0	−65.2	34.6	6.9	213.2	−42.6	ES
11/1	40 mT	7	6.639	199.6	−24.2	16.6	13.0	198.7	−74.2	SI
11/2a	60 mT	3	2.926	57.9	−21.5	27.0	15.6	57.2	19.1	ES
11/2b	60 mT	3	2.907	51.5	−46.2	21.6	17.4	41.2	−0.7	ES?
12										unstable
13	30 mT	8	7.923	20.9	4.5	90.5	5.2	4.5	24.5	WS
14	350°	9	8.613	178.4	6.5	20.7	10.3	177.4	20.9	?

15	300°	7	6.822	264.4	0.1	33.7	9.1	267.8	−22.1	ES
16	60 mT	10	8.970	60.1	15.1	8.7	15.0	70.9	−49.7	?
17	580°	14	13.752	124.5	13.9	52.3	5.2	18.5	18.5	ES
18	500°	10	9.818	241.9	4.2	49.5	6.3	242.3	−1.8	ES
19	400°	8	7.714	25.8	−16.5	24.5	10.0	30.0	−4.6	?
20/1	400°	3	2.976	88.1	1.7	81.8	8.9	95.3	12.3	WS?
20/2	400°	8	7.647	133.7	10.2	19.8	11.1	129.8	−18.8	WS
21	350°	12	11.880	97.1	34.2	91.4	4.2	124.0	8.3	WS
22	400°	8	7.304	258.0	−65.5	10.1	15.6	182.2	−6.0	WS
23	400°	9	8.767	9.1	59.9	34.4	8.0	280.8	72.8	SI
24/1	300°	4	3.751	230.6	−44.2	12.0	20.2	119.7	−79.4	SI
24/2	300°	7	6.950	261.9	−24.8	119.1	4.8	296.9	−68.7	SI
25	300°	8	7.796	344.5	62.8	34.3	8.4	192.3	71.0	SI
26	400°	11	10.695	240.1	24.8	32.8	7.4	240.2	−25.4	ES
27	400°	11	10.392	218.4	−42.5	16.4	10.4	232.5	−73.6	SI
28	400°	8	7.754	319.4	24.7	28.5	9.3	322.4	−49.6	?
29	400°	9	8.400	192.1	−7.2	13.3	12.8	194.0	17.2	ES?
30	400°	11	10.685	242.4	−64.4	31.8	7.5	299.6	−25.0	?
31										unstable
32										unstable
33	400°	6	5.908	184.1	62.5	54.5	7.7	86.3	66.0	SI
34	400°	10	9.597	155.5	58.1	22.3	9.4	345.5	61.7	SI
37	350°	16	15.907	196.0	−24.0	160.9	2.8	178.6	−33.9	karst region
38										unstable
40	500°	12	10.858	134.9	−8.5	9.6	13.0	134.3	10.8	WS?

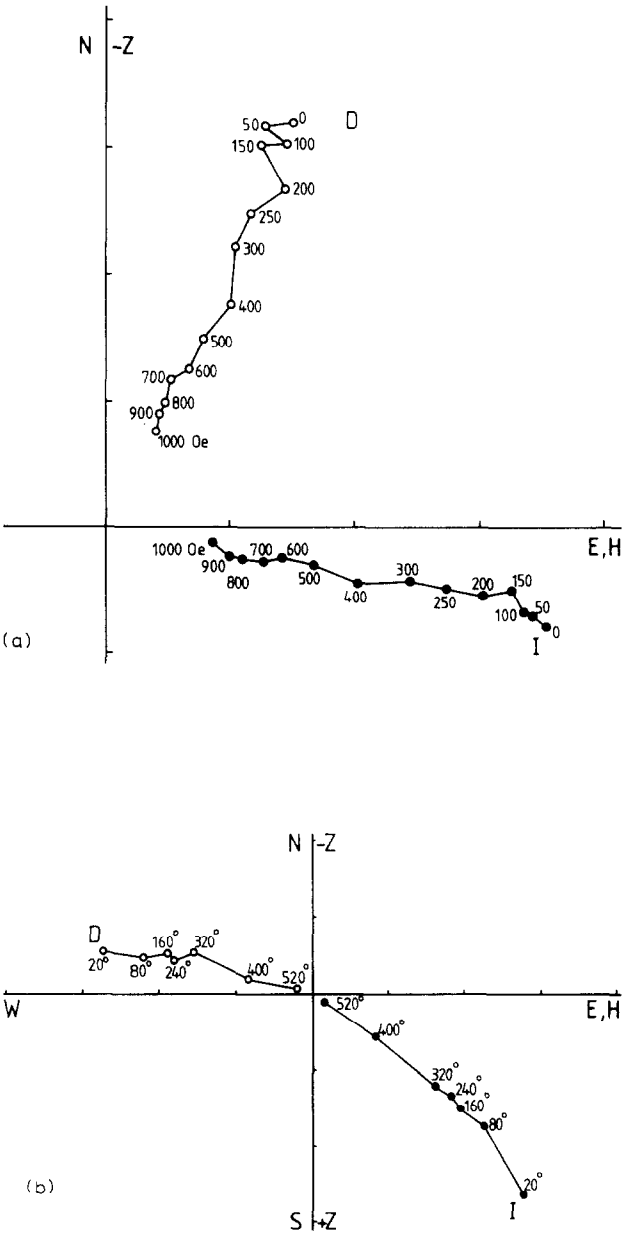


Fig. 5. Vector plots during demagnetization of NRM. Open symbols: projection into the horizontal plane. Closed symbols: Plot of the horizontal (H) versus the vertical component (Z). Steps are given either in °C or in Oe. (1 Oe $= 10^{-4}$ Tesla.)

a. AF demagnetization of a specimen from site number 13 (volcanic rock). 1 scale unit: 0.12 A/m.

b. Thermal demagnetization of a specimen from site number 21 (sandstone). 1 scale unit: $4 \cdot 10^{-3}$ A/m.

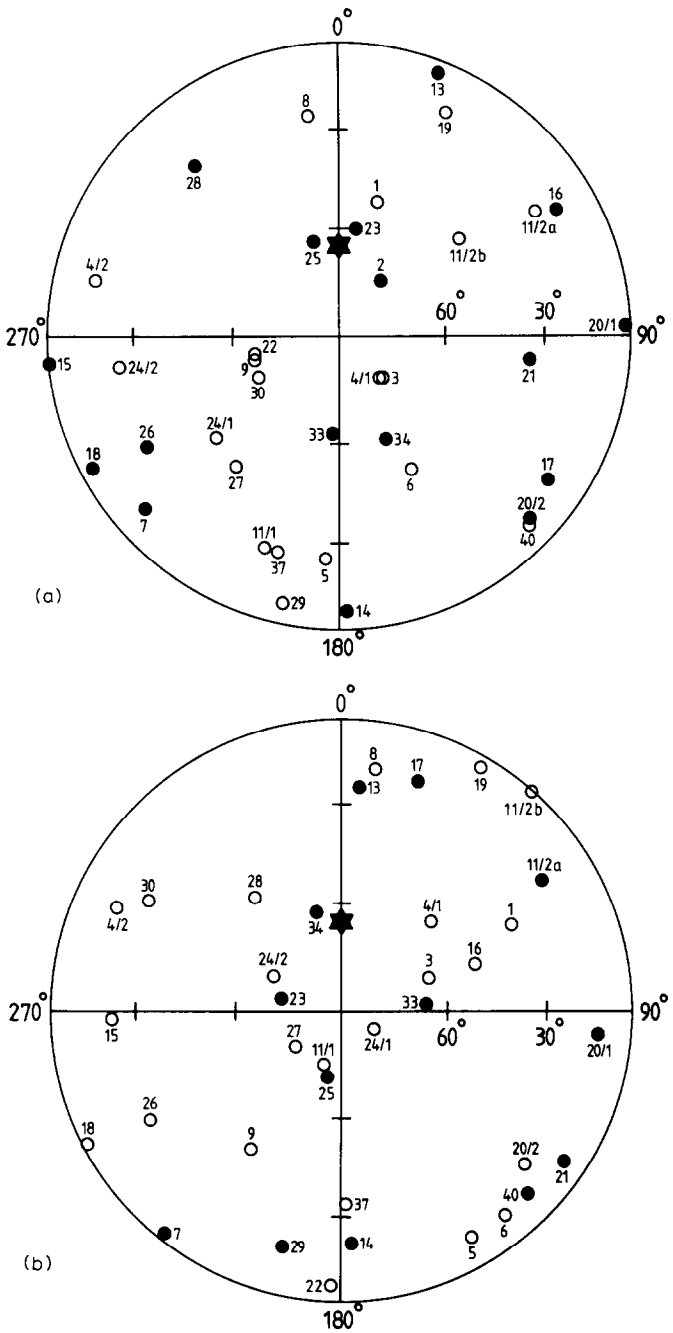


Fig. 6. Site mean directions of the ChRM. The ovals of confidence (see Table IV) are not plotted. Star: present geomagnetic field in the sampling area. Numbers refer to the site numbers. Open and closed symbols: negative or positive inclination, respectively. a. Before bedding corrections. b. After bedding corrections.

TABLE V

Sites with too steep inclinations (SI, see Table IV). Directions before and after bedding corrections, together with their means.

Site No.	Before bedding correction		After bedding correction	
	D (°E)	I (°)	$D_{corr.}$ (°E)	$I_{corr.}$ (°)
3	133.5	−73.1	69.5	−63.6
4/1	136.0	−73.8	45.6	−54.5
11/1	199.6	−24.2	198.7	−74.2
23 *	189.1	−59.9	100.8	−72.8
24/1	230.6	−44.2	119.7	−79.4
24/2	261.9	−24.8	296.9	−68.7
25 *	164.5	−62.8	12.3	−71.0
27	218.4	−42.5	232.5	−73.6
33 *	4.1	−62.5	266.3	−66.0
34 *	335.5	−58.1	165.0	−61.7
$N = 10$; $R = 8.220$;			$N = 10$; $R = 9.251$	
$D = 219.0°$; $I = -67.8°$			$D = 92.1°$; $I = -87.3°$	
$k_1 = 5.1$; $\alpha_{95} = 19.7°$			$k_2 = 12.0$; $\alpha_{95} = 12.8°$	

$k_2 : k_1 = 2.35$. Critical ratio for $N = 10 : 2.22$.
Fold test is positive at the 95% limit.

* Inverted directions.

The furnace could reach a temperature of 600°C (Schweitzer, 1975). In a later stage of the study, a thermal demagnetizer from Schonstedt Company (TSD-1) was used as well. AF demagnetization was carried out with an apparatus from Highmoor.

Figure 5a shows an example for an AF demagnetization of a specimen from site number 13, where a stable component could be isolated. A similar example for a successful thermal demagnetization is plotted in Fig. 5b (a specimen from site 21). The vector plots show the change of the vertical and horizontal components respectively in orthogonal projection.

From the vector plots appropriate optimum alternating fields and temperatures were chosen for the isolation of a stable and consistent characteristic remanence direction for each site.

The results are listed in Table IV and plotted in Fig. 6a (without the circles of confidence). The ChRM directions are very scattered. This scatter is not reduced visibly when the bedding corrections with the data of Table II are made. The corrected remanence directions are also listed in Table IV and plotted in Fig. 6b.

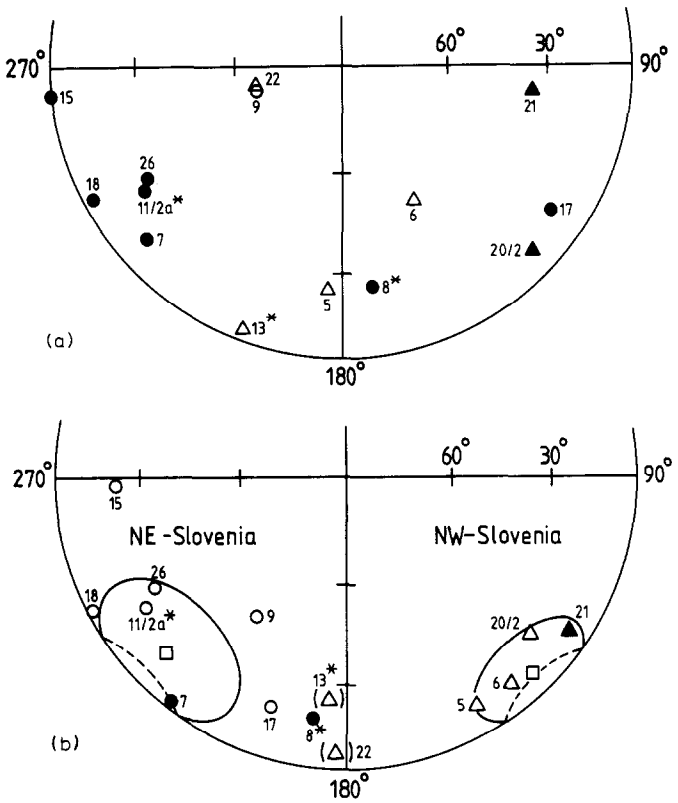


Fig. 7. Site mean directions of all sites having a ChRM with flat inclinations ($I < 45°$) after bedding corrections. Indexed with a star: projection to the other hemisphere. Open and closed symbols: negative or positive inclination respectively. Triangles: sites in northwestern Slovenia; circles: sites in northeastern Slovenia. a. Before bedding corrections. b. After bedding corrections. Triangles in brackets: see Table VI. Squares: mean directions for northwest or northeast Slovenia respectively, together with the ovals of 95% confidence.

INTERPRETATION OF THE ChRM DIRECTIONS

The ages of the investigated rocks range from Middle Permian to Middle Triassic. From the Apparent Polar Wander Path (APWP) of Europe (Irving, 1977) an average pole position of about 160°E, 46°N for the above time interval can be deduced. For the mean geographic coordinates of the sampling area (15°E, 46°N) a reversed palaeoremanence direction of $D = 204°$, $I = -14°$ can be expected. Only part of the sites (see Table IV) have the typical flat inclinations of ChRM after bedding correction, which must be expected from a remanent magnetization in this area of Permo-Triassic age. Many sites show steep inclinations, which are typical for much younger remanence ages (for instance the Tertiary). These sites with an inclination

TABLE VI

Sites from northwestern Slovenia (WS, see Table IV). Directions before and after bedding corrections together with the means.

Site No.	Before bedding correction		After bedding correction	
	D (°E)	I (°)	$D_{corr.}$ (°E)	$I_{corr.}$ (°)
5	183.7	−24.5	150.0	−10.9
6	151.4	−46.9	141.2	−10.6
13 *	200.9	−4.5	184.5	−24.5
20/2	133.7	10.2	129.8	−18.8
21	97.1	34.2	124.0	8.3
22	258.0	−65.5	182.2	−6.0
$N = 6$; $R = 3.737$			$N = 6$; $R = 5.431$	
$D = 162.6°$; $I = -21.9°$			$D = 151.4°$; $I = -11.4°$	
$k_1 = 2.2$; $\alpha_{95} = 38.5°$			$k_2 = 8.8$; $\alpha_{95} = 19.3°$	
$k_2 : k_1 = 4.0$. Critical ratio for $N = 6 : 2.97$. Fold test is positive at the 95% limit.				
Without sites number 13 and 22:				
$N = 4$; $R = 2.941$			$N = 4$; $R = 3.880$	
$D = 141.8°$; $I = -7.9°$			$D = 136.3°$; $I = -8.2°$	
$k_1 = 2.8$; $\alpha_{95} = 41.6°$			$k_2 = 25.0$; $\alpha_{95} = 14.0°$	
$k_2 : k_1 = 8.93$. Critical ratio for $N = 4 : 4.28$ (95% limit) and 8.47 (99% limit). Fold test is positive even at the 99% limit.				

* Inverted direction.

larger than 50° are labelled "SI" (Steep Inclination) in Table IV. Their fold test (according to McElhinny, 1964) is demonstrated in Table V. Although the ratio of the precision parameters after and before bedding correction is still indicative for a positive fold test, we regard the mean remanence direction after bedding correction as insignificant. It is almost vertical and even too steep for the Tertiary. We therefore consider the sites labelled with "SI" as being partially remagnetized during or after folding and tilting. It should also be noticed that these sites are predominantly situated either close to the Sava or close to the Idrija fracture zones respectively (see Fig. 2).

The sites with flat inclinations are shown in Fig. 7a before, and in Fig. 7b after bedding correction. When projected to one hemisphere, they are all located in the second and third quadrant respectively. Among these, a subdivision can be made between the sites in northeastern Slovenia (ES, circles) and those in northwestern Slovenia (WS, triangles), with a geographic dividing line in north–south direction near Ljubljana. The sites with flat inclinations in northwestern Slovenia (5, 6, 13, 20/2, 21, 22) have the typical NW–SE directed remanence directions, which have

TABLE VII

Sites from northeastern Slovenia (ES, see Table IV). Directions before and after bedding corrections, together with their means.

Site No.	Before bedding correction		After bedding correction	
	D (°E)	I (°)	$D_{corr.}$ (°E)	$I_{corr.}$ (°)
7	228.5	11.9	218.3	2.7
8 *	172.0	25.0	188.1	17.7
9	254.0	−65.2	213.2	−42.6
11/2a *	237.9	21.5	237.2	−19.1
15	264.4	0.1	267.8	−22.1
17	124.5	13.9	198.5	−18.5
18	241.9	4.2	242.3	−1.8
26	240.1	24.8	240.2	−25.4
$N = 8$; $R = 5.287$			$N = 8$; $R = 6.991$	
$D = 224.5°$; $I = 9.0°$			$D = 225.9°$; $I = -15.0°$	
$k_1 = 2.6$; $\alpha_{95} = 30.8°$			$k_2 = 7.0$; $\alpha_{95} = 18.8°$	

$k_2 : k_1 = 2.69$. Critical ratio for $N = 8 : 2.48$.
Fold test is positive at the 95% limit.

* Inverted directions.

also been found for the Permo-Triassic of the Italian part of the Southern Alps (Zijderveld and Van der Voo, 1973). The fold test is demonstrated in Table VI. It is definitely positive on the 95% limit, and even positive on the 99% limit, if only 4 of the best grouped sites are considered. The mean direction ($D = 151.4°$, $I = -11.4°$) is in agreement, within the limits of error, with the mean direction of the Bolzano Rhyolites (Zijderveld and Van der Voo, 1973) at $D = 148°$, $I = -20°$. The sites in northeastern Slovenia (7, 8, 9, 11/2a, 15, 17, 18, 26) show a mean direction of remanence ($D = 225.9°$, $I = -15.0°$) which is, within the limits of error, in agreement with the central European direction ($D = 204°$, $I = -14°$). The fold test is demonstrated in Table VII. It is positive on the 95% limit.

There are also sites from northwest and southeast Slovenia, where typical declinations for the corresponding areas have been found, but where the inclinations are still too far away from the more closely grouped directions so that they have not been included in the computation of a mean direction or in the fold test. They are labelled with "ES?" or "WS?" respectively. It is assumed that the remanence of these sites is still affected by the younger magnetic overprint.

ROCK-MAGNETIC INVESTIGATIONS AND POLISHED SECTION STUDIES

These studies were made to investigate the carrier of remanence and to demonstrate the effect of magnetic overprinting, which became evident from the analysis of the palaeomagnetic data.

Isothermal remanence acquisition measurements were made on the sandstones mainly in search for secondary goethite. The results are shown in Fig. 8. Only in some cases (curves of type A and B) the available field of 1.5 Tesla (15 kOe) turned out to be sufficient to show that hematite, eventually also smaller amounts of magnetite occur in the rocks, but definitely not goethite. For the sites with a curve of type C the presence of some goethite besides hematite cannot be excluded.

Many of the sandstones contain large amounts of detrital opaque minerals. Secondary mica (sericite) was visible even macroscopically. This indicates that the sandstones have suffered a small degree of regional metamorphism by which the primary content of ferrimagnetic ores was also affected. The occurrence of secondary hematite (pseudomorphic after magnetite) shows that the primary detrital remanence (DRM) with magnetite as carrier could have been later replaced at least partially by a secondary chemical remanence (CRM), carried by hematite and/or goethite.

Similar effects have been observed in the magmatic rocks. Those with the flat inclinations, typical for the Permo-Triassic, show the lowest degree of alteration by low-temperature oxidation. The most reliable data (site 13 in northwest Slovenia and site 11/2a in northeast Slovenia) come from sites with a primary high temperature oxidation (magnetite–ilmenite exsolution lamellae, needles of primary hematite in the silicates) and only few signs of a later low-temperature oxidation. The rocks of sites with steep inclinations do not exhibit any sign of a high temperature oxidation. Their primary ore phase has in general completely disappeared. These rocks possess only a very small secondary ore content (mostly hematite or maghemite) thus explaining well the unusually low NRM intensities. All rocks from the other sites show intermediate degrees of low temperature alteration with a more or less altered

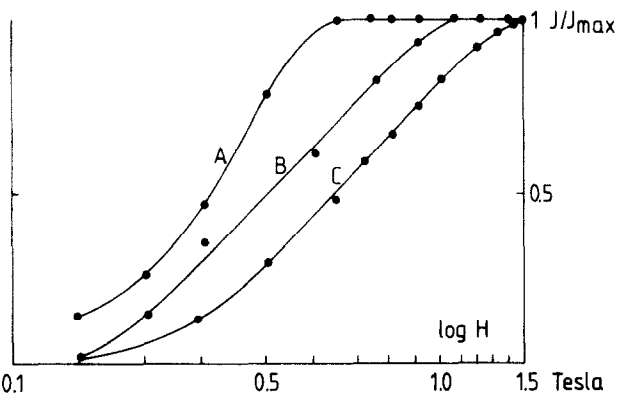


Fig. 8. Isothermal remanence acquisition curves for sandstones measured in fields up to 15 kOe (1.5 Tesla). Three types of behaviour have been found. Type *A*—sites: 28, 31, 32, 34; Type *B*—sites: 17, 18, 24, 26; Type *C*—sites: 19, 20, 21, 22, 23, 25, 27, 29, 30, 33. For interpretation see text. All remanence intensities have been normalized to the final intensity achieved with $H = 1.5$ Tesla.

primary ore content (titanomaghemites) plus some secondary minerals (hematite). This explains why the magnetic overprinting was total for many sites, but fortunately only partial for a sufficiently large number of other sites.

CONCLUSIONS

Figure 9 shows the geographic position of the sites listed in Tables VI and VII. All sites with unreliable data due to magnetic overprinting (steep inclinations, unstable remanence, uncertain tectonic position) have been omitted. Those with the typical "Adriatic" remanence directions are plotted as a triangle, those with typical "European" directions as circles respectively. The numbers are site numbers. It can be seen that the "Adriatic" directions can only be observed in the area west of Ljubljana. To the east of Ljubljana only the "European" directions occur. It is concluded from this regional distribution of the remanence directions that at least the northwestern part of Slovenia can still be regarded as a part of the Adriatic plate. The position of the northeastern margin of the Adriatic plate as proposed in a paper by Vandenberg (1979) seems to be correct in principle, but it should be shifted closer to the area of Ljubljana.

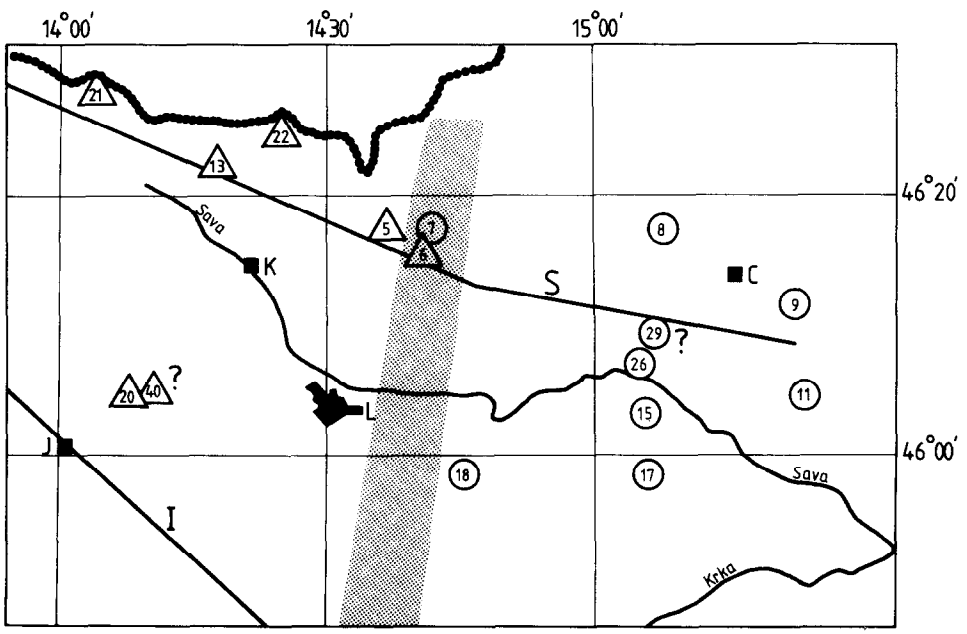


Fig. 9. Geographic distribution of the sites with typical "Adriatic" (triangles) and typical "European" (circles) remanence directions. Sites indexed by a "?" are those where the magnetic overprint could not entirely be removed. Other symbols: see legend of Fig. 2. Shaded area: proposed northeastern margin of the Adriatic plate.

This result is in conflict with that obtained by Márton and Márton (1981) from the Jurassic rocks of the Transdanubian Central Mountains in Hungary, where also typical "Adriatic" remanence directions have been determined. (Moreover, there are also geological arguments which do not support our palaeomagnetically determined plate boundary near Ljubljana. The main evidence stems from the Mesozoic sediments of the Slovenian trough with their continuous lithological and tectonic features between western and eastern Slovenia.)

More palaeomagnetic measurements are therefore needed in order to see whether the typical "European" directions in the area east of Ljubljana are just a local effect or whether the eastern margin of the Adriatic plate is in fact further in the east.

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